Chapter 6. Permutation Tests for Analysis of Variance

Kruskal–Wallis, ordered alternatives, Friedman test, general two-way layouts

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6.1 One-way layout

Motivation. To compare k groups without normality, rank all observations together and compare average ranks across groups; this is the Kruskal–Wallis test, the rank analogue of one-way ANOVA.

Observations $X_{ij} = \mu_i + \epsilon_{ij}$, $i = 1, \dots, k$, $j = 1, \dots, n_i$, $n = \sum n_i$, with i.i.d. errors. We test $H_0 : \mu_1 = \cdots = \mu_k$.

The F-test. With group means $\bar{X}_i$ and grand mean $\bar{X}$,

$$SS_{\text{Tr}} = \sum_i n_i(\bar{X}_i - \bar{X})^2, \qquad SS_E = \sum_{i,j}(X_{ij} - \bar{X}_i)^2, \qquad F = \frac{SS_{\text{Tr}}/(k-1)}{SS_E/(n-k)}.$$

Proposition 6.1

$SS_{\text{Tot}} := \sum_{i,j}(X_{ij} - \bar{X})^2 = SS_{\text{Tr}} + SS_E$. Under H_0 with normal errors, $F \sim F_{k-1,n-k}$.

Proof [Supplement]. Expand $X_{ij} - \bar{X} = (X_{ij} - \bar{X}_i) + (\bar{X}_i - \bar{X})$. The cross term is $2\sum_i(\bar{X}_i - \bar{X})\sum_j(X_{ij} - \bar{X}_i) = 0$. Under normality, $SS_E/\sigma^2 \sim \chi^2_{n-k}$ is independent of the group means. SS_{Tr}/σ^2 is the squared length of the projection of the standard normal vector $(\sqrt{n_i}(\bar{X}_i - \mu)/\sigma)_i$ onto the complement of $(\sqrt{n_i})_i$, hence χ^2_{k-1}. ■

Permutation F-test. Under H_0 (equal distributions) all n observations are exchangeable, so relabel them among the groups: there are $n!/\prod n_i!$ equally likely splits. Since SS_{Tot} is fixed, $F = \frac{n-k}{k-1} \cdot \frac{SS_{\text{Tr}}}{SS_{\text{Tot}} - SS_{\text{Tr}}}$ is increasing in SS_{Tr}, and both give the same test. For moderately large samples the permutation distribution of F is close to $F_{k-1,n-k}$ even without normality.

6.1.1 The Kruskal–Wallis test

Rank all observations jointly. Let $\bar{R}_i$ be the mean rank of group i and S_R^2 the sample variance of all ranks ($= n(n+1)/12$ without ties).

Definition (Kruskal–Wallis statistic)

$KW = \dfrac{1}{S_R^2}\sum_{i=1}^k n_i\Big(\bar{R}_i - \dfrac{n+1}{2}\Big)^2$, which without ties equals $\dfrac{12}{n(n+1)}\sum_i n_i(\bar{R}_i - \frac{n+1}{2})^2$.

Proposition 6.2

(i) The permutation test based on KW coincides with the permutation F-test applied to the ranks.
(ii) Under H_0, $KW \xrightarrow{d} \chi^2_{k-1}$ as all $n_i \to \infty$.

Proof. (i) For ranks, $SS^R_{\text{Tot}} = (n-1)S_R^2$ is fixed and $SS^R_{\text{Tr}} = S_R^2\, KW$, so $F_R = \frac{n-k}{k-1}\frac{KW}{n-1-KW}$ is increasing in KW.

(ii) With $Z_i = \sqrt{n_i}(\bar{R}_i - \bar{R})/S_R$, the without-replacement covariances (Theorem 4.2) give $\text{Cov}(Z) \to I - \mathbf{u}\mathbf{u}'$ with $\mathbf{u} = (\sqrt{n_i/n})_i$, a rank-$(k-1)$ projection. A multivariate finite-population CLT makes Z asymptotically $N(0, I - \mathbf{u}\mathbf{u}')$, so $KW = \|Z\|^2 \xrightarrow{d} \chi^2_{k-1}$. ■

In small samples the χ^2 approximation can be poor in the tails, so the exact or Monte Carlo permutation distribution is preferred.

6.1.2 Ordered alternatives

Against $H_1 : \mu_1 \le \mu_2 \le \cdots \le \mu_k$ (at least one strict), the **Jonckheere–Terpstra** statistic sums the pairwise Mann–Whitney counts:

$$JT = \sum_{i<l} U_{il}, \qquad U_{il} = \#\{(a,b) : X_{ia} < X_{lb}\}.$$

Large JT supports the trend. It equals (up to an affine map) Kendall's τ between the observations and the group index, and its permutation distribution gives an exact test. Using the order information makes it more powerful than KW against monotone trends.

6.2 Randomized complete block design

k treatments in each of b blocks, one observation per cell: $X_{ij} = \mu + t_i + \beta_j + \epsilon_{ij}$. We test $H_0 : t_1 = \cdots = t_k$.

Classical F. $SS_{\text{Tr}} = b\sum_i(\bar{X}_{i\cdot} - \bar{X})^2$ and $SS_E = \sum_{i,j}(X_{ij} - \bar{X}_{i\cdot} - \bar{X}_{\cdot j} + \bar{X})^2$. Under normality, $F = \frac{SS_{\text{Tr}}/(k-1)}{SS_E/((k-1)(b-1))} \sim F_{k-1,(k-1)(b-1)}$.

Why shuffle within blocks. Under H_0, $X_{ij} = \mu + \beta_j + \epsilon_{ij}$: observations in the same block are exchangeable, but observations in different blocks are not (their means differ by block effects). The permutation distribution therefore permutes treatment labels independently within each block, giving $(k!)^b$ equally likely arrangements. Within-block shuffles leave the block means and SS_{Tot} unchanged. Hence $SS_E = SS_{\text{Tot}} - SS_{\text{Blocks}} - SS_{\text{Tr}}$ moves opposite to SS_{Tr}, and the permutation F-test is the same as the permutation test based on SS_{Tr}.

6.2.1 The Friedman test

Rank the k observations within each block. Let $R_i = \sum_j R_{ij}$ be the rank sum of treatment i and $\bar{R}_{i\cdot} = R_i/b$.

Definition (Friedman statistic)

$$FM = \frac{12b}{k(k+1)} \sum_{i=1}^{k} \left(\bar{R}_{i\cdot} - \frac{k+1}{2}\right)^2 = \frac{12}{bk(k+1)} \sum_i R_i^2 - 3b(k+1) \text{ (no ties).}$$

Under H_0 the within-block rank vectors are independent uniform permutations, so the null distribution is distribution-free, and $FM \xrightarrow{d} \chi^2_{k-1}$ as $b \to \infty$. With ties, divide by the average within-block rank variance instead of $k(k+1)/12$.

Example 6.1

$k = 3$, $b = 4$, within-block ranks of (A, B, C): $(1,2,3)$, $(1,3,2)$, $(1,2,3)$, $(2,1,3)$. The rank sums are $5, 8, 11$, so $FM = \frac{12}{48}(25 + 64 + 121) - 48 = 4.5$. The χ^2_2 approximation gives $p = e^{-2.25} = 0.105$. Enumerating all $(3!)^4 = 1296$ arrangements gives the exact $p = 162/1296 = 0.125$. With few blocks the exact distribution is noticeably different.

6.2.2 Page's trend test

For $H_1 : t_1 \le \cdots \le t_k$, use $L = \sum_{i=1}^{k} iR_i$, a sum over blocks of a Spearman-type statistic between treatment order and within-block ranks.

Proposition 6.3

Under H_0, $E(L) = \dfrac{bk(k+1)^2}{4}$ and $\mathrm{Var}(L) = \dfrac{bk^2(k+1)(k^2-1)}{144}$.

Proof. Within one block, $L_j = \sum_i iR_{ij}$ with $(R_{1j}, \ldots, R_{kj})$ a uniform permutation. Each R_{ij} has mean $\frac{k+1}{2}$, so $EL_j = \frac{k+1}{2} \cdot \frac{k(k+1)}{2}$. By Proposition 5.2 applied to $a_i = i$ and $b =$ ranks, $\mathrm{Var}\, L_j = \frac{1}{k-1}\Big(\sum_i (i - \frac{k+1}{2})^2\Big)^2 = \frac{1}{k-1}\Big(\frac{k(k^2-1)}{12}\Big)^2 = \frac{k^2(k+1)(k^2-1)}{144}$. The blocks are independent; multiply by b. ■

6.3 General two-way layouts

With c replicates per cell and no interaction, $X_{ijl} = \mu + t_i + \beta_j + \epsilon_{ijl}$. The permutation test shuffles all kc observations within each block. The rank version (**Skillings–Mack**) ranks within blocks and uses

$$SM = \frac{b\sum_i (\bar{R}_{i\cdot} - \bar{R})^2}{\text{average within-block rank variance}} \dot{\sim} \chi^2_{k-1}.$$

Again, the permutation distribution is more reliable than the χ^2 approximation in small designs. Caution: when interaction is present, a nonlinear transformation such as ranking can destroy or create interaction. Then use special aligned-rank procedures, or treat the model as a regression and use the resampling methods of Chapter 10.

6.4 Selected exercises

Motivation. The exercises connect the Friedman test with the sign test and with Cochran's Q for binary data, and extend the null moments of Page's statistic to tied ranks.

Exercise 6.5 (Friedman with two treatments is the sign test)

In a complete block design with $k = 2$ treatments, b blocks and no ties within blocks, show that the permutation test based on the Friedman statistic FM is the two-sided sign test applied to the differences $D_j = X_{1j} - X_{2j}$.

Solution.

Step 1. Let $S = \#\{j : D_j > 0\}$. Within block j treatment 1 has rank $R_{1j} = 1 + I(D_j > 0)$, so $R_1 = b + S$ and $R_2 = 3b - R_1 = 2b - S$.

Step 2. By the second form of the definition with $k = 2$,

$$\begin{aligned} FM &= \frac{2}{b}\{(b+S)^2 + (2b-S)^2\} - 9b = \frac{2}{b}(5b^2 - 2bS + 2S^2) - 9b \\ &= \frac{(2S-b)^2}{b} = \Big(\frac{S - b/2}{\sqrt{b/4}}\Big)^2. \end{aligned}$$

So FM is an increasing function of $|S - b/2|$, and it is the square of the standardized sign statistic.

Step 3. The within-block permutations swap the two labels in each block independently with probability $\frac{1}{2}$, which flips the sign of D_j. Hence under the permutation distribution $S \sim B(b, \frac{1}{2})$, the null law of the sign statistic (Proposition 2.1). Rejecting for large FM is rejecting for large $|S - b/2|$, the two-sided sign test. The χ^2_1 approximation to FM is the normal approximation to S.

Consequently the Friedman test inherits the efficiency of the sign test: for $k = 2$ and normal errors its ARE relative to the F-test (here the paired t-test) is $2/\pi$ (Table in Section 2.3). The general value $3k/(\pi(k+1))$ (a known result, not proved here) increases toward $3/\pi$ as k grows. ■

Exercise 6.6 (Page's L with ties)

In a complete block design rank within blocks using mid-ranks, and let S_j^2 be the sample variance (divisor $k-1$) of the ranks in block j. Show that under H_0

$$E(L) = \frac{bk(k+1)^2}{4}, \qquad \mathrm{Var}(L) = \frac{k(k-1)(k+1)}{12} \sum_{j=1}^{b} S_j^2,$$

and that this reduces to Proposition 6.3 without ties.

Solution.

Step 1. Under H_0 the observations within a block are exchangeable, so given the multiset of mid-ranks in block j, the vector $(R_{1j}, \ldots, R_{kj})$ is a uniform random permutation of it. Different blocks are independent.

Step 2. Mid-ranks keep the total $k(k+1)/2$, so each R_{ij} has mean $\frac{k+1}{2}$ and $E(L_j) = \frac{k+1}{2} \sum_i i = \frac{k(k+1)^2}{4}$, where $L_j = \sum_i i R_{ij}$.

Step 3. The proof of Proposition 5.2 uses only that fixed numbers $b_1, \ldots, b_k$ are randomly permuted against fixed $a_1, \ldots, a_k$. It gives $\mathrm{Var}\left(\sum_i a_i b_{\pi(i)}\right) = S_{aa} S_{bb}/(k-1)$, where $S_{aa} = \sum_i (a_i - \bar{a})^2$, after removing the constant $\bar{a} \sum_i b_i$. Take $a_i = i$, so $S_{aa} = k(k^2-1)/12$, and $b =$ the mid-ranks of block j, so $S_{bb} = (k-1)S_j^2$. Then

$$\mathrm{Var}(L_j) = \frac{k(k^2-1)}{12} S_j^2.$$

Step 4. Summing over independent blocks gives the stated mean and variance. Without ties $S_j^2 = k(k+1)/12$, and $\mathrm{Var}(L) = \frac{k(k-1)(k+1)}{12} \cdot \frac{bk(k+1)}{12} = \frac{bk^2(k+1)(k^2-1)}{144}$. ■

Exercise 6.7 (Binary responses: Cochran's Q)

Let $x_{ij} \in \{0, 1\}$ (success or failure of treatment i in block j), with treatment totals T_i, block totals B_j and grand total T. The tie-corrected Friedman statistic is $FM = b \sum_i (\bar{R}_{i\cdot} - \frac{k+1}{2})^2 / (\frac{1}{b} \sum_j S_j^2)$, with mid-ranks and S_j^2 as in Exercise 6.6.
(a) Show that FM equals Cochran's statistic

$$Q = \frac{(k-1)(k \sum_i T_i^2 - T^2)}{kT - \sum_j B_j^2}.$$

(b) Four treatments of plantar warts were applied to each of 7 patients (blocks), with outcomes (1 = success):

	P1	P2	P3	P4	P5	P6	P7
Freezing	0	1	0	0	0	1	0
Surgery	1	1	0	1	0	1	1
Topical medication	0	0	0	1	0	1	0
Duct tape	1	1	0	0	1	0	1

Test whether the treatments differ.

Solution.

(a)

Step 1. (Mid-ranks.) In block j the $k - B_j$ failures share positions $1, \dots, k - B_j$, with mid-rank $c_j = \frac{k-B_j+1}{2}$. The successes share the remaining positions, with mid-rank $\frac{2k-B_j+1}{2} = c_j + \frac{k}{2}$. So $R_{ij} = c_j + \frac{k}{2}x_{ij}$; this also holds when $B_j \in \{0, k\}$.

Step 2. (Numerator.) Averaging over blocks, $\bar{R}_{i\cdot} = \bar{c} + \frac{k}{2b}T_i$. The average over i of $\bar{R}_{i\cdot}$ is $\frac{k+1}{2}$, so $\bar{R}_{i\cdot} - \frac{k+1}{2} = \frac{k}{2b}(T_i - T/k)$ and

$$b\sum_i \Big(\bar{R}_{i\cdot} - \frac{k+1}{2}\Big)^2 = \frac{k^2}{4b}\Big(\sum_i T_i^2 - \frac{T^2}{k}\Big).$$

Step 3. (Denominator.) $S_j^2 = \frac{k^2}{4} \cdot \frac{1}{k-1}\sum_i \big(x_{ij} - \frac{B_j}{k}\big)^2 = \frac{k^2}{4(k-1)}\Big(B_j - \frac{B_j^2}{k}\Big) = \frac{kB_j(k-B_j)}{4(k-1)}$.

Step 4. (Ratio.) Using $\sum_j B_j(k - B_j) = kT - \sum_j B_j^2$,

$$FM = \frac{\frac{k^2}{4b}\big(\sum_i T_i^2 - T^2/k\big)}{\frac{1}{b} \cdot \frac{k}{4(k-1)}\big(kT - \sum_j B_j^2\big)} = \frac{(k-1)\big(k\sum_i T_i^2 - T^2\big)}{kT - \sum_j B_j^2} = Q.$$

Blocks with $B_j \in \{0, k\}$ contribute nothing to either part: they carry no information about treatment differences.

(b) $T_i = (2, 5, 2, 4)$, $T = 13$, $\sum T_i^2 = 49$; $B_j = (2, 3, 0, 2, 1, 3, 2)$, $\sum B_j^2 = 31$. So

$$Q = \frac{3(4 \cdot 49 - 169)}{4 \cdot 13 - 31} = \frac{81}{21} = 3.86.$$

The χ^2_3 approximation gives $p = 0.28$. Enumerating all 13,824 within-block arrangements of the observed 0's and 1's gives the exact permutation p-value $5148/13824 = 0.37$. Either way there is no evidence of a difference between the treatments. With only six informative blocks the χ^2 approximation is noticeably off, as in Example 6.1. ■